

CvSU Vision

The premier university in historic Cavite globally recognized for excellence in character development, academics, research, innovation and sustainable community engagement.

**CAVITE STATE UNIVERSITY**

Bacoor City Campus
Soldiers Hills IV, Molino VI,
City of Bacoor, Cavite
☎ (046) 476 - 5029
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CvSU Mission

Cavite State University shall provide excellent, equitable and relevant educational opportunities in the arts, science and technology through quality instruction and responsive research and development activities.

It shall produce professional, skilled and morally upright individuals for global competitiveness.

CropCare Weather-Driven Strategies and**Management****Solutions for Sustainable Agriculture**

By:

Aaron Gabriel Arlegue -50

Kevin Joshnel Mesina -50

Mark Angelo Barnido -50

Mike Anthony Oliva -50

Jam Ryan Nazareno -50

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CHAPTER I SHOULD BE IN CENTERED

Welcome to CropCare, your reliable source of comprehensive and understandable crop information. Our platforms' dedication is to give farmers and business/land owners with the latest update and detailed advice from your fellow Farmers, enthusiast or business owners.

Let me take you to a little trip to share and enhance your knowledge about farming.

DESCRIPTIVE FORMAL

The rationale of the study, which focuses on the pressing need for sustainable farming practices, aligns seamlessly with the objective of sharing knowledge, experiences, and connections among fellow farm enthusiasts. As we strive into researching sustainable agricultural methods, our study's primary aim is to address the challenges posed by environmental concerns and resource limitations. Simultaneously, Our goal of fostering a community where farmers can exchange insights, learn from each other's experiences, and establish vital connections. By investigating sustainable practices, we not only aim to contribute to agricultural advancement but also to create a collaborative space that empowers individuals within the farming community to collectively innovate and thrive in an ever-evolving agricultural landscape.

AVOID USING

FIRST POV

INSTEAD USE

THIRD PERSON

POINT OF VIEW

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In our country, incorporating weather-driven strategies into sustainable farming practices presents a complex challenge. However, the specific influence of these weather-responsive approaches on farmers' adoption of sustainable farming methods and their impact on knowledge sharing remains uncertain. This study seeks to investigate how the use of weather-responsive strategies affects the application of sustainable farming methods. Furthermore, it aims to explore how these strategies impact the exchange of knowledge within the agricultural sector.

The essence of addressing the problem has several reasons why:

- **Enhance sustainability** - Understanding the impact of weather-driven strategies on sustainable farming methods enables the development of more resilient agricultural practices. This, in turn, promotes environmental sustainability and resource efficiency in the face of changing weather patterns and climate variability.
- **Enhances Farming Practices** - Insights gained from this research can help farmers optimize their practices by aligning them with weather-driven strategies

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and the knowledge that they learn from fellow farmers from different agricultural sectors.

- **Knowledge Growth and Community Engagement** - It encourages collaboration, the sharing of best practices, and the dissemination of vital information among farmers, facilitating collective learning and growth within certain sectors.
- **Economic and Social Impact** - Improved farming practices can have a direct positive impact on the economic well-being of farmers and the communities reliant on agriculture. Sustainable and weather-adaptive techniques can potentially increase incomes, reduce risks, and contribute to food security.

Our Solution is Technology adoption, Community engagement and Education, With CropCare you can get these three solution in one website. Technology Adaptation can help or assist the farm enthusiast and farmers by informing them about the decision that aligned with the weather conditions. Community Engagement with the collaboration and knowledge-sharing events within the farming community. A culture of sharing experiences and successful practices related to weather-responsive farming. Education Implementing educational initiatives and training sessions that focus on weather-responsive agricultural practices can empower farmers with the knowledge and skills necessary to adapt to changing weather patterns effectively.

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In closing the gap of weather-driven strategies, sustainable farming practices, and knowledge exchange among farmers, these kind of efforts can make way for a resilient, informed, and interconnected agricultural community. By embracing innovation, education, collaboration, we envision a future where farmers thrive amidst changing weather patterns, empowered by shared knowledge and adaptive practices, ensuring not just sustainable yields but also the sustainability of our agricultural landscapes.

Section II

This thesis explores the integration of optimized crop management, improved crop selection as essential components of sustainable agricultural practices. By examining their multifunctional roles in resource utilization, soil health, and climate resilience, the study aims to establish the significance of these integrated practices in enhancing agricultural sustainability and productivity.

This provides an overview of the multifaceted functions of optimized crop management, and improved crop selection in sustainable crop production, emphasizing the importance of integrating these practices for achieving agricultural efficiency and resilience.



• Optimized Crop Management

water and drought management

The role of optimized crop management, focusing on efficient resource utilization, soil health and fertility maintenance, pest and disease control, climate resilience, and economic viability. This includes precision farming, data-driven decision making, and the use of innovative tools to monitor and optimize crop growth, yield, and quality. By implementing this, farmers can maximize their resources and minimize waste, resulting in higher productivity and profitability.

• Improved Crop Selection

The chapter discusses the importance of improved crop selection in enhancing agricultural sustainability, resilience, and economic viability. It uses modern breeding techniques, genetic engineering, and trait selection to develop crops more resilient to climate change, pests, and diseases. Farmers can maximize yields and quality while reducing chemical input demand by choosing enhanced crop types.

Our Project stands out for several reasons:

Remove this or rephrase

The project aims to offer a user-friendly platform with insights from farmers, enthusiasts, and business owners for essential crop-related information. It informs users



of the most recent advancements while building a sense of community and shared knowledge within the agricultural sector. The platform ensures accessibility and benefits everyone, even for those who are new to farming or crop management. By combining the knowledge gap between theory and practice, the project hopes to improve crop cultivation's sustainability and overall success.

The Uniqueness of our project:

The platform offers comprehensive crop information sourced from the expertise of farmers, enthusiasts, and business owners, utilizing real-time data to provide current insights on crop management techniques.

One example of the project's originality, innovation, and significance is its use of farmer-provided, real-time collected data, which offers insightful and up-to-date information on crop management techniques. A valuable and cutting-edge resource in the agricultural sector, the platform's emphasis on thorough and understandable information also guarantees accessibility for farmers and business/landowners.

The Beneficiaries and their Benefits they could get from

The beneficiaries of this study are farmers, agricultural businesses, and landowners, who can gain practical, timely, and reliable insights into crop management.

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By leveraging the information provided by the study, they will benefit from enhanced decision-making capabilities, improved crop yields, and optimized resource allocation, ultimately leading to increased profitability and sustainability in their agricultural practices.

Section III

- CropCares' objective is to revolutionized agriculture by prioritizing intelligent farming decision, sustainable farming practices, Community engagement. With the existence of many certain innovative products that seeks to safeguard crops against diseases, pests, and weeds while maintaining high-quality outputs. CropCare focuses on ensuring the long-term viability of agriculture while also supporting farmers through education, weather-index initiatives, and strong community engagement. This comprehensive support system fosters economic viability and encourages sustainable practices, directly addressing essential environmental needs.

The researchers specifically aim:

1. To design CropCare with the following

Features:

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- a. Capable of giving daily weather update depending on the location provided
- b. Capable of interacting to fellow farmer/business owner
- c. Capable of giving a strong community engagement

2. To Develop CropCare using JavaScript Frameworks.**FrontEnd**

- React JS

BackEnd

- Firebase

Database

- Firebase

3. To test CropCare using Alpha and Beta Testing procedures.**4. To evaluate CropCare using ISO/IEC 25010:2011 Software Evaluation Tool for I.T.**

Experts and Actual Users.

5. To implement CropCare in Agricultural Sectors Specifically Farmers and Farm lot owners

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Section IV

Scope

- The scope of CropCare encompasses a holistic approach to agricultural sustainability, focusing on various key elements such as farmer education, weather-index initiatives, and robust community engagement.

Limitation

- The limitations within CropCare's scope may revolve around resource accessibility, technological adoption in certain regions, and potential cultural or socio-economic barriers affecting the implementation of sustainable practices and community engagement initiatives.

- 1.) Improve the UI/UX of the website
- 2.) Improve a proper logo
- 3.) Subrang gulo tignan ng mga elements ng website
di matanto kung webbased or mobile based
- 4.) Ang pangat ng font choices
- 5.) Icons go to flat icon website
- 6.) Much better magkaroon ng mapa ng phil.
yung parang sa weather reporting
- 7.) Yung "report" sa Daily Forecast:
 - Sana paayos yung design
 - Font choices
 - Font color theory
- 8.) Use proper dummy data

- login logo / login page
- React.js why is that the UI is lack of design
- spacing in account summary view
- font
- spacing
-